

भारतीय सूचना प्रौद्योगिकी संस्थान धारवाड़

Indian Institute of Information Technology Dharwad (IIIT Dharwad)

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Course Project — Problem Statement

EC466 — Software Defined Networks

Session: Aug – Nov 2026 · Credits: 3-1-0-0-4

SELF-X: A Self-Improving Multi-Agent Large Language Model (LLM) Framework for Autonomous Software-Defined Network (SDN) Flow Management

Problem Statement 1

Self-Improving Multi-Agent LLM Framework for Autonomous SDN Flow Management

Category: Agentic AI for SDN Flow Control

Course & Evaluation Details

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Team ID	SDN_07
Project Theme	SELF-X

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Companion: SELF-X Architecture (Fig. 1) & Self-Improvement Loop (Fig. 2)

1. Abstract

Modern software-defined networks (SDNs) centralise intelligence in a controller that must continuously install, update and delete flow rules across a large number of OpenFlow switches. As traffic becomes bursty, dynamic and application-driven, static rule engineering and hand-tuned policies fail to keep latency, throughput and energy in balance. Existing machine-learning controllers (deep reinforcement learning, DRL) retrain on fixed reward functions and remain rigid when the traffic mix or topology changes. The 2026 research landscape — NetSoft 2026 under the theme 'Autonomous and Reliable Softwarized Networks in the Age of Distributed Intelligence', NFV-SDN 2026, and the GLOBECOM 2026 'Agentic AI' workshop — identifies agentic and self-adaptive intelligence as the key enabler for next-generation autonomous networks.

In this problem statement we address the question: **A centralised SDN controller must make flow-management decisions that satisfy multiple, often conflicting objectives (low latency, high throughput, bounded packet loss, low energy and resource cost) while reacting to rapidly changing traffic. Controllers today rely on hand-crafted policies that are brittle, and single-learner DRL agents that cannot generalise or self-correct when the reward signal, topology or workload drifts. How can a controller autonomously learn, decompose and refine its own flow-management policy over time — without human re-engineering — while guaranteeing safety and explainability of every installed rule?**

2. Proposed System Architecture

We propose SELF-X: a self-improving multi-agent LLM framework layered above an SDN controller. An LLM oracle planner decomposes the global intent (e.g. 'minimise average latency under SLA') into sub-tasks handled by specialist agents (flow-sizing, routing/engineering, anomaly, security, slicing). Each agent selects and invokes controller tools (OpenFlow/Flow-Mod, restconf, stats pull) through a tool gateway. A reward and reflection loop observes the network state after each action, computes a multi-objective reward and returns it to an experience-replay buffer. The system self-improves by (i) self-critique of failed actions, (ii) reward-guided selection among LLM prompts/policies, and (iii) periodic lightweight self-fine-tuning (LoRA) of the policy model using high-reward episodes.

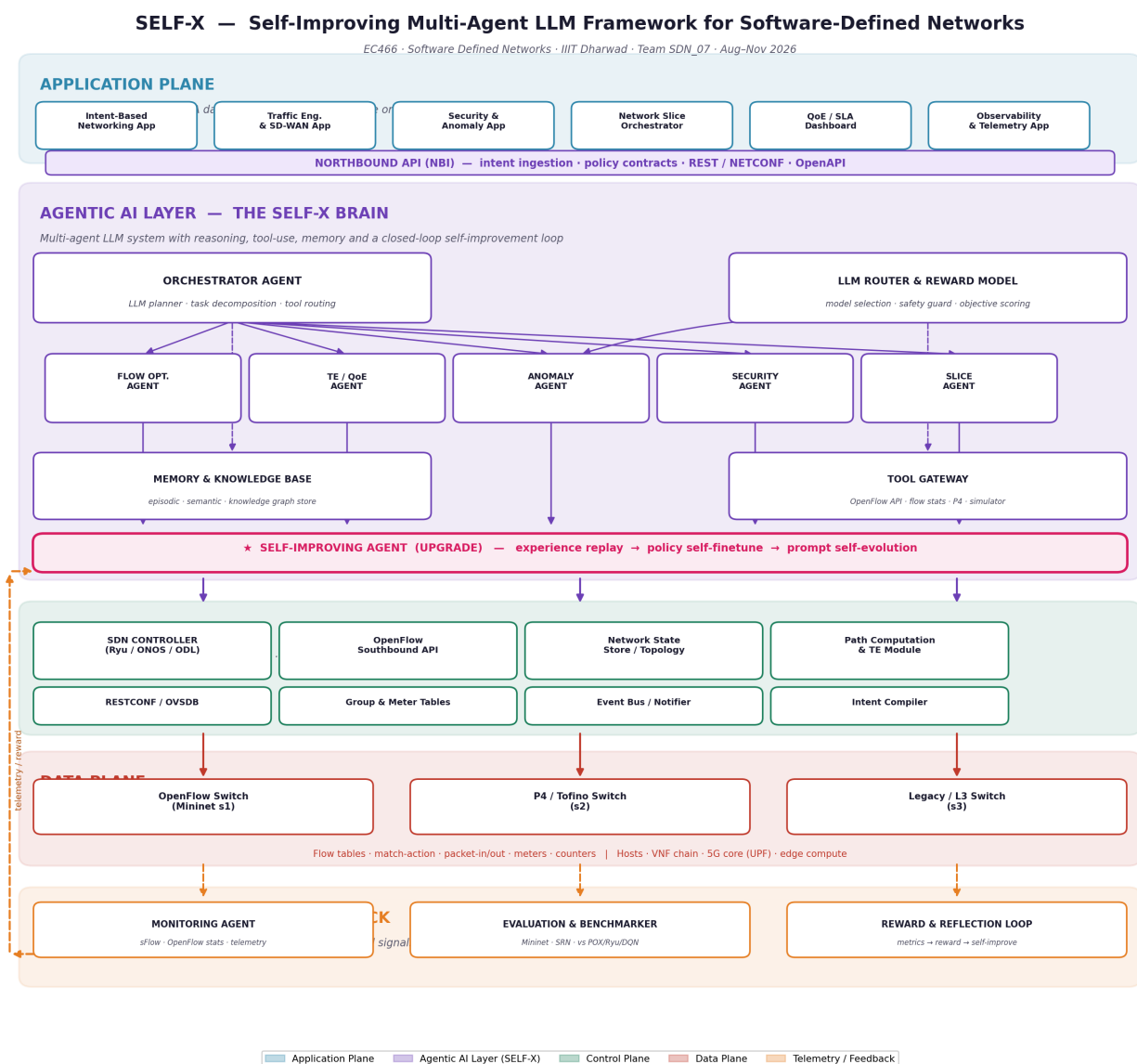


Figure 1 — SELF-X: self-improving multi-agent LLM framework for SDN flow management (architecture).

See Figure 1 (SELF-X architecture). The architecture exposes four planes: an Application plane of intent/policy apps that talk to the Agentic AI layer over a northbound API; the Agentic AI layer (orchestrator + five specialist agents + memory/knowledge base + tool gateway + the self-improving agent); the SDN Control plane (Ryu/ONOS, southbound OpenFlow, state store, path computation); and the Data plane (OpenFlow/P4 switches in Mininet). A Telemetry plane closes the loop by feeding monitoring, evaluation and reward back into the Agentic AI layer.

3. Problem Statement

PROBLEM:

A centralised SDN controller must make flow-management decisions that satisfy multiple, often conflicting objectives (low latency, high throughput, bounded packet loss, low energy and resource cost) while reacting to rapidly changing traffic. Controllers today rely on hand-crafted policies that are brittle, and single-learner DRL agents that cannot generalise or self-correct when the reward signal, topology or workload drifts. How can a controller autonomously learn, decompose and refine its own flow-management policy over time — without human re-engineering — while guaranteeing safety and explainability of every installed rule?

4. Motivation / Need

Modern software-defined networks (SDNs) centralise intelligence in a controller that must continuously install, update and delete flow rules across a large number of OpenFlow switches. As traffic becomes bursty, dynamic and application-driven, static rule engineering and hand-tuned policies fail to keep latency, throughput and energy in balance. Existing machine-learning controllers (deep reinforcement learning, DRL) retrain on fixed reward functions and remain rigid when the traffic mix or topology changes. The 2026 research landscape — NetSoft 2026 under the theme 'Autonomous and Reliable Softwarized Networks in the Age of Distributed Intelligence', NFV-SDN 2026, and the GLOBECOM 2026 'Agentic AI' workshop — identifies agentic and self-adaptive intelligence as the key enabler for next-generation autonomous networks.

Keywords: multi-agent LLM, self-improvement, OpenFlow, flow management, autonomous networking

5. Objectives

1. Design and implement a multi-agent LLM orchestrator that decomposes high-level intents into controller actions for SDN flow management.
2. Provide a tool gateway that lets agents safely read/write flows, meters, groups and stats through the southbound API.
3. Implement a reward-and-reflection loop that scores agent actions on latency, throughput, loss, and resource cost.
4. Add a genuine self-improvement mechanism (self-critique + replay + LoRA fine-tuning) and demonstrate that the agent's policy improves with experience.
5. Prove the framework works on an emulated multi-switch Mininet topology.

6. Proposed Methodology / Approach

- Build Mininet topologies (linear/leaf-spine) with Open vSwitch and OpenFlow 1.3.
- Select Ryu (and ONOS) as the controller to receive the SELF-X agent layer.
- Define a flow-management task set: elephant flow rerouting, QoS/rate limiting, congestion avoidance, and anomaly-driven rule flush.
- Implement specialist agents using an LLM backend with ReAct-style tool calling over the OpenFlow/RESTCONF gateway.

- Instrument a reward signal (queue + port + flow stats) and an experience replay memory.
- Enable self-improvement: self-critique on negative rewards, prompt/policy selection, and LoRA fine-tuning on high-reward trajectories.
- Benchmark against baseline Ryu, POX, a static-flow controller and a DQN controller.

7. Expected Deliverables

- SELF-X multi-agent codebase (orchestrator, 5 agents, tool gateway, reward loop).
- Mininet topologies and traffic generator (iperf3, hping3).
- Self-improvement module (replay buffer, critic, LoRA fine-tuner).
- Benchmark harness producing latency/throughput/loss/energy tables and figures.
- This problem statement and the accompanying submission-ready report and presentation.

8. Evaluation Metrics

- Average end-to-end latency, throughput (Mbps), packet loss (%), flow-install convergence time.
- Controller CPU / control-channel overhead and number of packet-in messages.
- Reward convergence and number of self-improvement iterations to reach target quality.
- Generalisation: performance under unseen topologies and bursty workloads.

9. Novelty / Originality

Integration of a genuine closed-loop self-improvement mechanism (self-critique + experience replay + on-policy LLM fine-tuning) with multi-agent SDN control, instead of a static DRL network or a stateless prompt-based agent. This directly targets NetSoft 2026's 'agentic AI and multi-agent systems' and 'self-organising, self-healing, self-optimising softwarized networks' themes.

10. Benchmarking Against Existing Systems

We compare SELF-X against the following representative baselines to establish that agentic, self-improving control is justified:

S.No	Baseline / System Compared Against
1	Baseline 1: standard Ryu / POX controller with static OpenFlow rules.
2	Baseline 2: a Deep Q-Network (DQN) controller trained on the same reward.
3	Baseline 3: a single (non-specialist) LLM agent without the self-improvement loop.
4	Comparison metric: quality-of-service and learning efficiency / convergence time.

Comparison is performed on a common Mininet topology and workload, using the metrics in Section 8. The benchmarking harness is itself a deliverable (see Problem Statement 8).

11. Self-Improving Agent (Upgrade)

SELF-X is not a static prompt-based agent. It is augmented with an explicit self-improvement unit:

- Self-critique: after each action, a critic LLM reflects on the difference between intended and observed outcomes and produces an improved reasoning trace.
- Experience replay: a prioritized buffer stores (state, action, reward, reflection) episodes so the agent can learn from both successes and failures.
- On-policy self-finetuning: periodically, the underlying LLM policy is fine-tuned (LoRA) on high-reward trajectories, so the policy itself — not just the prompt — improves.
- Prompt/policy selection: the agent dynamically routes among candidate prompt-policies by their empirical reward, adapting to a drifting workload without human re-engineering.

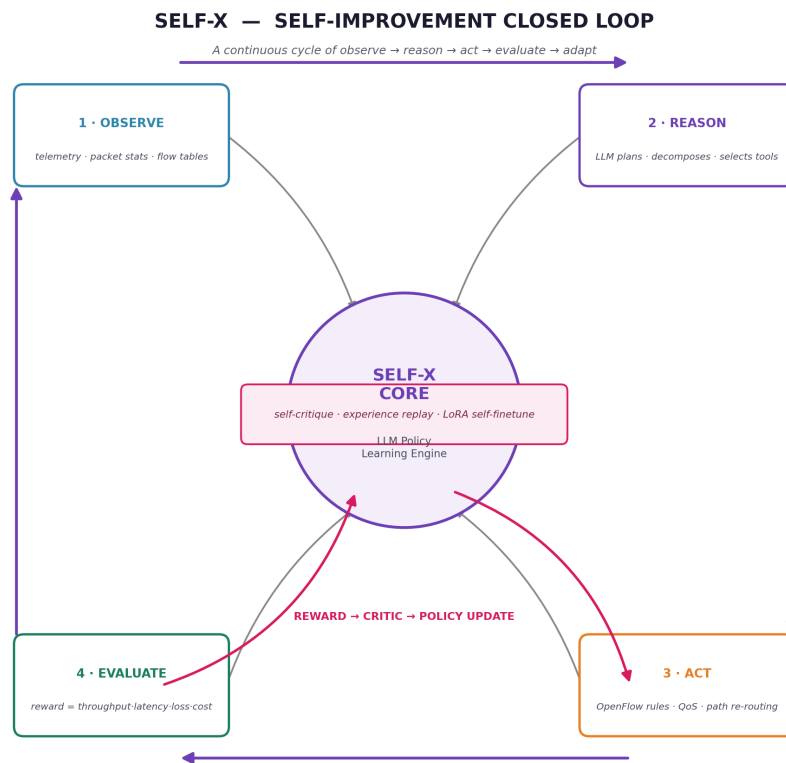


Figure 2 — SELF-X closed-loop self-improvement cycle (observe → reason → act → evaluate → adapt).

12. Expected Outcome / Impact

A demonstrably self-improving SDN controller whose multi-agent LLM policy lowers end-to-end latency and loss while adapting to unseen traffic — a concrete instance of an agentic, self-optimising softwareized network.

13. Alignment with 2026 Research Frontiers

This work targets the 2026 research frontier of autonomous and agentic-empowered softwareized networks, as framed by: **NetSoft 2026** — 'Autonomous and Reliable Softwareized Networks in the Age of Distributed Intelligence', whose scope explicitly lists Agentic AI and Multi-Agent Systems, Generative AI, explainable and trustworthy AI for network autonomy, and self-organising/self-healing/self-optimising softwareized networks; **IEEE NFV-SDN 2026**, which calls for agent-based SDN/NFV, AI/ML network automation and self-healing operations; and the **GLOBECOM 2026 / ICC 2026** workshops on Agentic AI-empowered wireless communications and networking, Network Slicing and O-RAN/edge convergence. By combining multi-agent LLM reasoning with a genuinely self-improving controller, SELF-X addresses the scalability,

interpretability and adaptability gaps that currently limit autonomous SDN.

14. Conclusion

This problem statement defines SELF-X, a self-improving multi-agent LLM framework that autonomously manages SDN flows while continuously learning from its own decisions. The project is compelling in 2026 because it sits exactly at the convergence of two active research thrusts — agentic AI and softwarized/autonomous networks — and contributes an architecture, a reproducible benchmark, and a concrete self-improvement mechanism that the community can reuse and extend.

15. References

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— End of Problem Statement —